

Kyle Macasilli-Tan

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EDUCATION

University of Southern California	Los Angeles, CA
B.S. in Computer Science	Expected May 2027
M.S. in Computer Science (Progressive Degree)	Expected Dec 2027
GPA: 3.46 Presidential Scholar (half-tuition merit scholarship)	
Relevant Coursework: Artificial Intelligence, Data Structures, Object-Oriented Design, Computer Systems, Computer & Network Security, Linear Algebra	

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, JavaScript, SQL, HTML/CSS
AI / ML: PyTorch, TensorFlow, LLMs, RAG, vector databases, prompt engineering, neural networks, computer vision
Tools & Platforms: React, Node.js, FastAPI, Git, Docker, Qdrant, MuJoCo, Google Cloud, REST APIs

EXPERIENCE

SLURM (Robotic Manipulation) Lab — USC	Feb – Aug 2025
<i>Project Lead & Researcher</i>	Los Angeles, CA
- Led testing of a PPO reinforcement-learning policy for complex bimanual manipulation tasks, reaching ~80% average success rate - Built a bimanual Franka robot-arm model in MuJoCo to perform cooperative manipulation tasks - Developed a hand-teleoperation system to accelerate data collection and bootstrap imitation learning	

Kroger	Jun – Aug 2024
<i>Software Engineering Intern</i>	Cincinnati, OH
- Optimized an LLM prototype using retrieval-augmented generation (RAG) to help associates retrieve analytics and resources, achieving an 81% successful-response rate - Automated tag-attribute assignment across 1,400+ files with conditional RegEx, improving component reusability and accessibility - Built loading placeholders and skeleton states, reducing Cumulative Layout Shift (CLS) by 0.05	

Integrity Express Logistics	Summers 2021 & 2022
<i>Software Engineering Intern</i>	Cincinnati, OH
- Built a load-tracking tool covering 500,000+ loads annually, supporting a ~62% revenue increase - Led migration of 23% of the main application from PHP to TypeScript, reducing bundle size and improving load times - Developed a React app used by 500+ employees nationwide and a shipment portal serving 3,000+ partner companies (e.g., Gordon Food Service, Del Monte)	

PROJECTS

Docs Navigator — Python, FastAPI, Qdrant, BGE-M3, Claude	2026
Live: docs-navigator.onrender.com · Code: github.com/kylemtan/docs-navigator	
- Built and deployed a multi-library RAG documentation assistant (Next.js, LangChain, LlamaIndex, Haystack) returning source-cited answers with ~97% LLM-judged faithfulness; live on Render - Designed a 70-question benchmark and eval harness (recall@k, MRR, faithfulness, out-of-scope refusal); improved retrieval recall@3 from 75% to 88% (page) and 58% to 73% (section) via BGE-M3 hybrid search and a domain-matched cross-encoder reranker - Ran controlled ablations across dense, hybrid, and reranking configs, finding reranker-embedder model-family alignment drove the largest retrieval gain	
SempAI — React, TypeScript, Node.js, Claude API, Tavily API, Kroger API	2026
Live: semp-ai.onrender.com · Code: github.com/kylemtan/SempAI	
- Built an AI meal-planning agent using Claude and the Tavily API to generate personalized meal plans from user dietary goals and constraints - Integrated the Kroger API to auto-populate a shopping cart and schedule grocery delivery, creating an end-to-end plan-to-delivery workflow	

ACTIVITIES

Center for AI in Society (CAIS++) — USC	Aug 2024 – Present
<i>Member</i>	
- Selected for USC’s premier undergraduate AI program (30-member cohort from 300+ applicants) - Built a CNN with batch normalization and max pooling to classify diabetic-retinopathy severity, reaching 73.77% accuracy	